# Statistical Analysis of Infant Birth Weight

Turning early risk into early action — protecting newborns and strengthening healthcare system


Group 8

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# Introduction

## Premature birth rates in US remain at historic high, report finds

**Rate in 2023 remains at 10.4% due to chronic conditions, inadequate prenatal care and racial disparities**

Premature birth rate in 2023 at 10.4%, while rate 10 years ago was 9.6%. Photograph: Zoonar GmbH/Alamy

Babies born too early or too small have higher chances of health problems in childhood and adulthood.

World Health Organization

Newborn Health

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## Preterm and low birth weight infants

**Preterm infants are born at less than 37 weeks gestational age and low birth weight infants are born with a birth weight below 2.5kg regardless of gestational age**. An estimated 15 million newborns are born preterm and more than 20 million are born low birth weight each year. Prematurity and low birth weight remains the leading cause of death in newborns and children under-five years. Preterm and LBW infants have a higher risk of developmental disabilities including cerebral palsy and retinopathy of prematurity. The consequences of prematurity and low birth weight may continue into adulthood, increasing the risk of adult onset chronic conditions such as obesity and diabetes.

Credits +

- The project studies what affects a baby's weight at birth.
- Low birth weight can cause serious health problems.
- We focus on the mother's health, habits, and background.
- The goal is to find key factors and improve prenatal care.

*Source: https://www.theguardian.com/us-news/2024/nov/14/premature-birth-rates-historic-high
https://www.who.int/teams/maternal-newborn-child-adolescent-health-and-ageing/newborn-health/preterm-and-low-birth-weight

# The Problem We've Always Known

- We have long known that low birth weight is one of the strongest indicators of a newborn's health.
- It is closely linked to infant mortality, cognitive delays, and lifelong health complications.
- According to the WHO, over 20 million babies are born underweight every year.
- Despite decades of research and prenatal care improvements, the problem persists.
- We're still asking: which pregnancies are most at risk — and why?

TIME

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**The trend is getting worse**

The preterm birth rate is now at its highest point since 2014. One of the main drivers of this is that food insecurity in America has also risen to its highest point over that same period.

In 2022, 1 in 12 babies in America was born with a low birthweight. This rate puts the U.S. in a similar ranking with Hungary, Turkey, and Brazil. Nordic countries, on the other hand, have half the rate of low birthweights as is seen in the U.S., largely due to strong healthcare systems, low pollution, healthier foods, and higher incomes.

**Over the last 30 years, more babies are born with low birthweights**

*Source:https://time.com/6965173/americas-birthweight-crisis/

# The Gap Between Data and Action

- Despite advances in prenatal care, early warning signs for low birth weight still go unnoticed.
- Current tools are too broad, missing the nuances of individual pregnancies.
- This gap between what we know and what we act on is where preventable harm occurs.

News Releases | Apr 17, 2025

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## New ACOG Guidance Recommends Transformation to U.S. Prenatal Care Delivery

**Washington, D.C.**—The American College of Obstetricians and Gynecologists is recommending significant changes to the way prenatal care is delivered in the United States, according to clinical guidance released today. The new, tailored, patient-centered approach could improve outcomes for millions of pregnant patients.

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MANAGEMENT AND ADMINISTRATION

## Florida Invests Millions to Keep Babies Alive. It's Not Working

Last year the state spent more than $170 million to address maternal and infant death, yet rates of infant and fetal mortality, as well as preterm and low-weight births, haven't improved much since a decade ago.

*Source: https://www.acog.org/news/news-releases/2025/04/new-acog-guidance-recommends-transformation-to-us-prenatal-care-delivery
https://www.governing.com/management-and-administration/florida-invests-millions-to-keep-babies-alive-its-not-working

# Understanding Our Dataset

- **Data Source:** Kaggle – Fall 2024 Birth Weight Prediction Competition (Over 90,000 birth records from a large observational study)

**Key Variables Collected:**

- Maternal Factors: age, education, weight gain, prenatal visits
- Paternal Factors: age, education, race/ethnicity
- Behavioral Factors: smoking, alcohol use during pregnancy
- Medical History: hypertension, diabetes, eclampsia, previous infant outcomes
- Demographics: race, marital status, Hispanic origin

**Target Variable:**

- Low Birth Weight (LBW) :
  - 1 = birth weight < 5.5 lbs
  - 0 = healthy weight

**Data Preparation Steps:**

- Handled missing values
- Removed outliers from key numeric predictors
- Encoded categorical variables into interpretable features

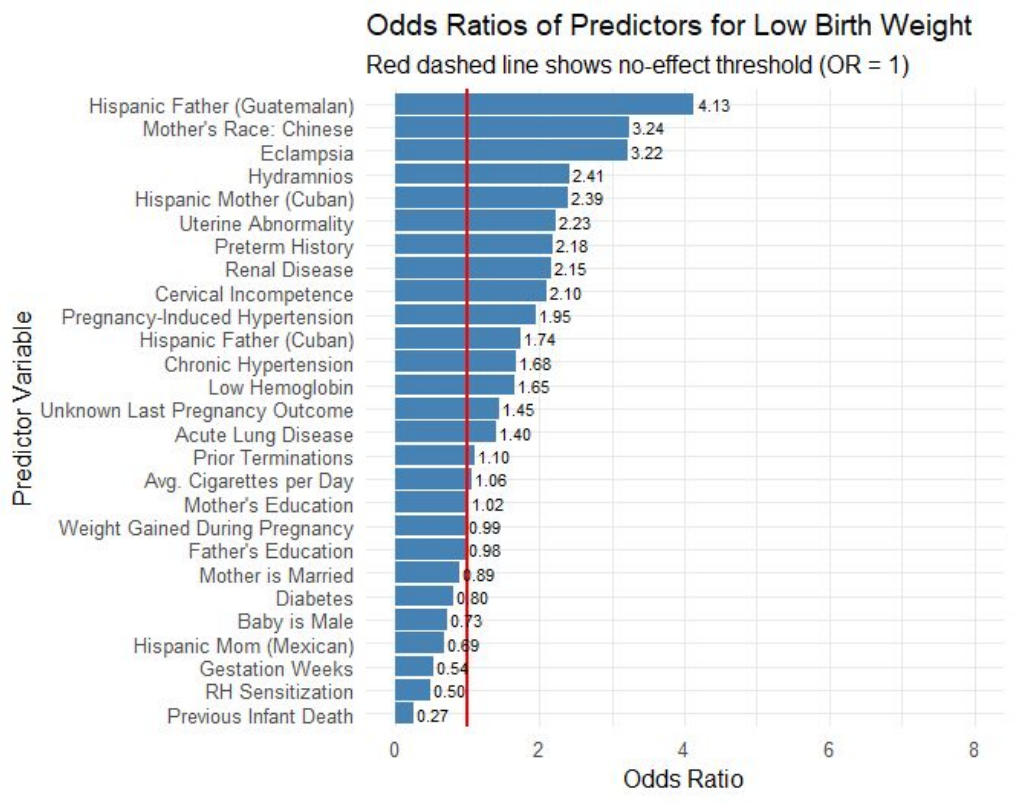
# Modeling Approach and Rationale

## Model Chosen: Logistic Regression

### Why Logistic Regression?

- **Perfect for Binary Outcomes:**
  Predicts Low Birth Weight (yes = 1, no = 0)
- **Clear Interpretability:**
  Coefficients explain how each factor impacts LBW risk
- **Odds Ratios:**
  Quantify the effect size of each predictor
- **Significance Testing:**
  Identifies statistically meaningful predictors via p-values
- **Policy and Clinical Relevance:**
  Supports targeted interventions and evidence-based healthcare policies

# Key Findings from Our Analysis



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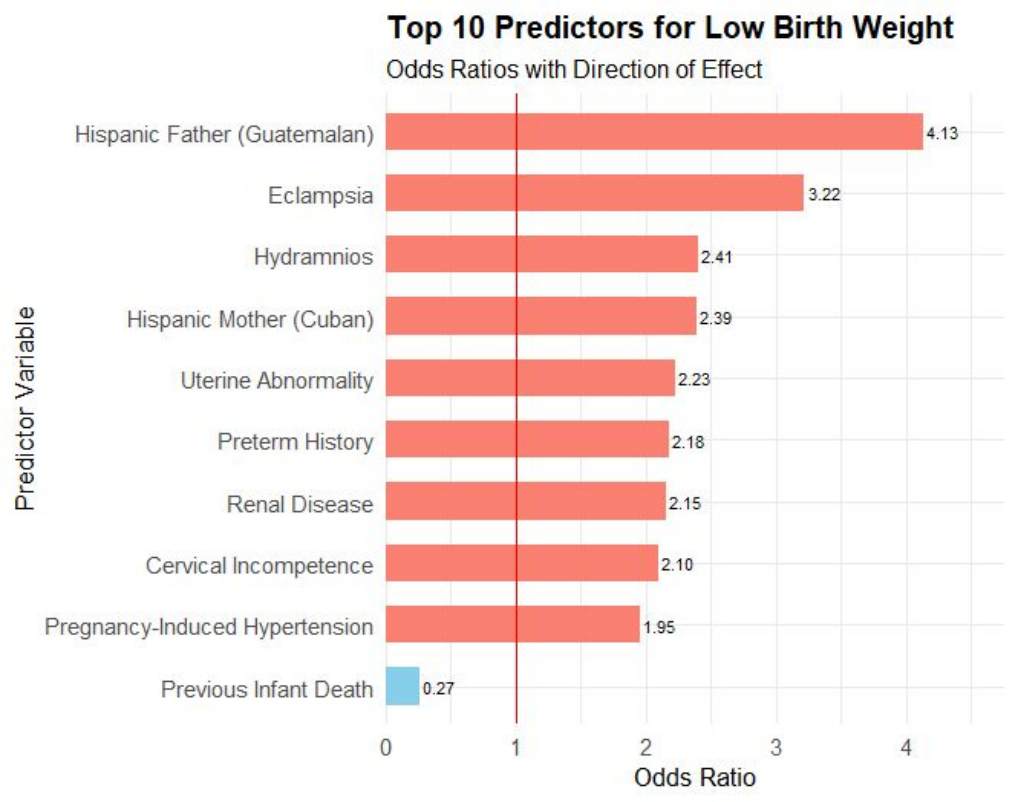


- Babies born to Guatemalan fathers were much more likely to have low birth weight compared to non-hispanic fathers.
- Medical problems like eclampsia, hydramnios, uterus problems, kidney disease, made low birth weight much more likely.
- Surprisingly, mothers who had lost a baby before were less likely to have a low birth weight baby this time, possibly because they received closer medical care.

# Summary

**Healthcare policymakers:**

- **Develop culturally tailored education programs** for Guatemalan fathers and Cuban mothers, especially in regions with **high concentrations of these communities**.

**Hospital Administrators:**

- **Integrate risk flags and referral prompts** into the Electronic Medical Record (EMR) system.
- **Allocate** staff and resources more efficiently.
- **Early risk identification** helps hospitals intervene sooner, prevent NICU cases, improve outcomes, and strengthen quality scores linked to reputation and Medicaid performance.

# THANK YOU!